Title: Simplifying filling out payment information using Google Pay or Apple Pay.

Start date: //_____

Finish date: //_____

Responsible: Yuliia Stepanova

Description

Test goal: increase adding payment information
Hypothesis: The development team has added the ability for users to fill in payment information using Google Pay or Apple Pay. The team believes that this simplification will increase the `add_payment_info` / `session` by 2% and, as a result, increase the percentage of start of payment (`begin_checkout` / `session`) by 2%. Importantly, the `begin_checkout` event can also occur without `add_payment_info`.

Options

Option A (Control): without changes.
Option B (Experimental): users with the option to add payment information via Google Pay or Apple Pay.

Data sampling

Target audience: all site visitors.
Traffic split: 50% for option A, 50% for option B

Metrics

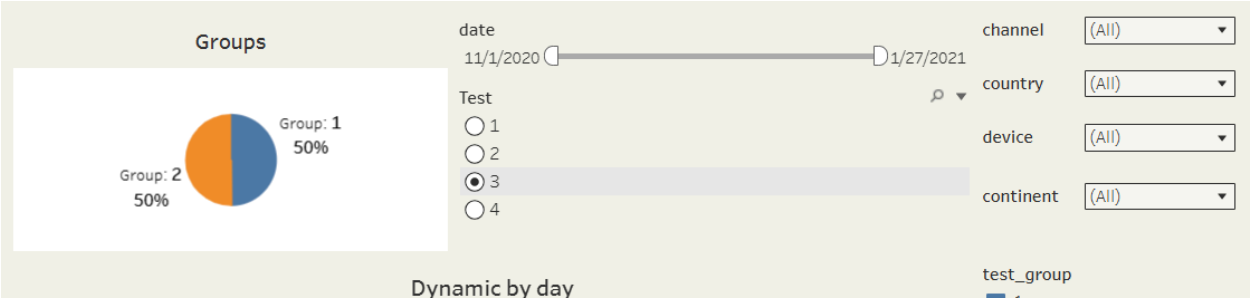
Primary metric: `add_payment_info` / `session`, growth of 2%.
Additional metrics: `begin_checkout` / `session`, growth of 2%.

Results

	value		value (%)
	test_group		test_group
event_name	1	2	2
add_payment_info	3,623	3,697	2.043%
add_shipping_info	5,298	5,188	-2.076%
add_to_cart	17,674	17,229	-2.518%
begin_checkout	9,532	9,264	-2.812%
first_visit	50,438	50,728	0.575%
new account	5,856	5,822	-0.581%
page_view	286,351	284,287	-0.721%
scroll	110,360	109,564	-0.721%
select_promotion	2,020	1,996	-1.188%
session	70,047	70,439	0.560%
session with orders	6,951	7,151	2.877%
session_start	71,312	71,686	0.524%
user_engagement	249,921	248,325	-0.639%
view_item	93,931	92,841	-1.160%
view_promotion	41,169	40,959	-0.510%

Analyze

Checking the 50% to 50% distribution:



Change in key metrics: In total, the add_payment_info / session metric increases by 1.5% and this is not a statistically significant result.

	test_group		
event_name	1	2	
add_payment_info	3623	3697	
session	70047	70439	
	5,17%	5,25%	1,5%

Evan's Awesome A/B Tools (home):

Sample Size Calculator | **Chi-Squared Test** | Sequential Sampling | 2 Sample T-Test | Survival Times | Count Data

Question: Does the rate of success differ across two groups?

	# successes	# trials	Confidence interval	[clear] [link]
Sample 1:	3623	70047		*5% – 5.3%*
Sample 2:	3697	70439		*5.1% – 5.4%*

Verdict:

No significant difference

(p = 0.52)

Confidence level: 95%

Let's check how much time we need to wait for the result to become statistically significant:

Evan's Awesome A/B Tools (home):

Sample Size Calculator | Chi-Squared Test | Sequential Sampling | 2 Sample T-Test | Survival Times | Count Data

Question: How many subjects are needed for an A/B test?

Baseline conversion rate:	5.17 %	*5.17%*	[link]
Minimum Detectable Effect:	1.5 %	*5.09% – 5.25%*	

The Minimum Detectable Effect is the smallest effect that will be detected (1-β)% of the time.

○ Absolute
◉ Relative

Conversion rates in the gray area will not be distinguishable from the baseline.

Sample size:

1,282,427

per variation

We see that it is necessary to have a very large number (1282k) of sessions in each of the groups. Now we have about 70k. That is, we still need to expect 19 similar ranges (1282427/70000 = 18.32) along with what is currently participating in the test.

Because the `begin_checkout` event can occur without `add_payment_info` **let's check the additional metric**:

	test_group			
event_name	1	2		
begin_checkout	9535	9264		
session	70047	70439		
	13,61%	13,15%	-3,4%	

Evan's Awesome A/B Tools (home):

Sample Size Calculator | **Chi-Squared Test** | Sequential Sampling | 2 Sample T-Test | Survival Times | Count Data

Question: Does the rate of success differ across two groups?

	# successes		# trials	Confidence interval	[clear] [link]
Sample 1:	9532	/	70047	13.4% – 13.9%	
Sample 2:	9264	/	70439	12.9% – 13.4%	

Verdict:

Sample 1 is more successful

(p = 0.012)

Confidence level: 95%

We see that this indicator did not increase, but instead fell by 3.4%, and the result is statistically significant. That is, the implemented changes negatively affect user behavior in general.

Let's check by segments, on which it is possible to implement a new method of filling in payment information.

or Undefined channels, there is a statistically significant positive change in the main and additional metrics:

	value		value (%)
	test_group		test_group
event_name	1	2	2
add_payment_info	492	542	10.16%
add_shipping_info	587	655	11.58%
add_to_cart	1,808	1,747	-3.37%
begin_checkout	1,066	1,175	10.23%
first_visit	89	87	-2.25%
new account	383	363	-5.22%
page_view	21,118	20,495	-2.95%
scroll	9,687	9,455	-2.39%
select_promotion	124	115	-7.26%
session	4,636	4,534	-2.20%
session with orders	443	494	11.51%
session_start	4,523	4,398	-2.76%
user_engagement	19,605	19,015	-3.01%
view_item	7,802	7,524	-3.56%

Evan's Awesome A/B Tools (home):

Sample Size Calculator | **Chi-Squared Test** | Sequential Sampling | 2 Sample T-Test | Survival Times | Count Data

Question: Does the rate of success differ across two groups?

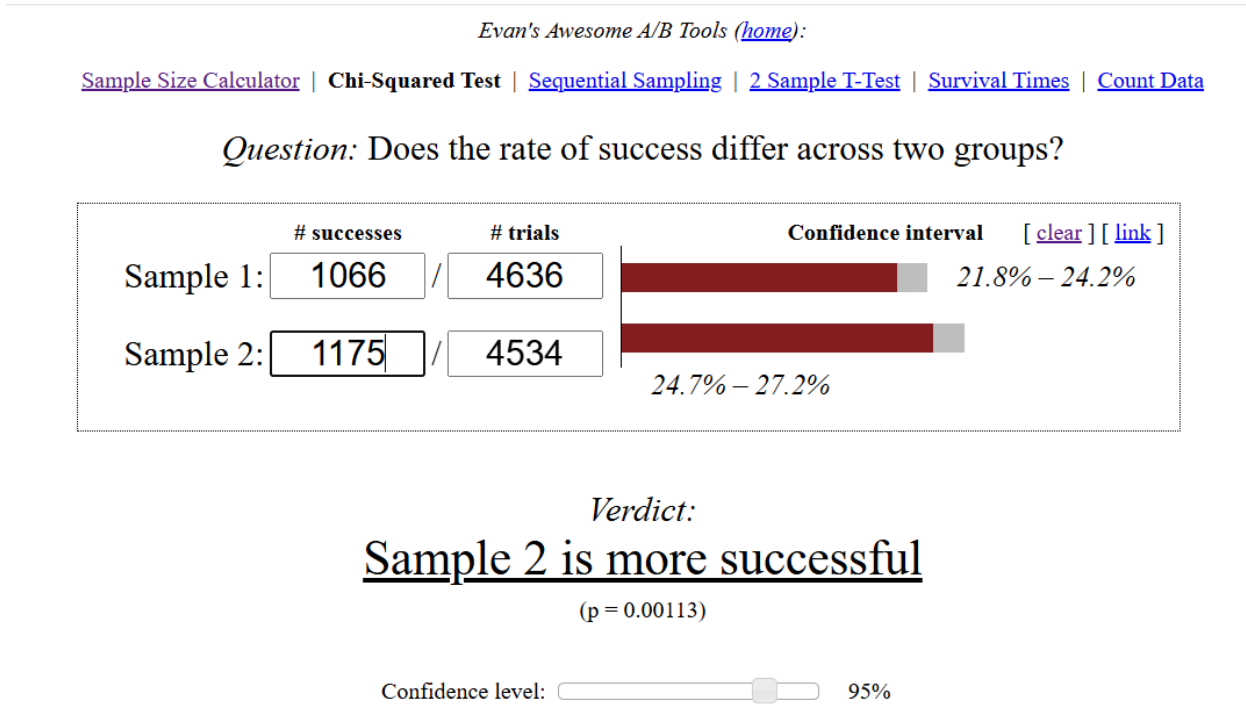
	# successes		# trials	Confidence interval	[clear] [link]
Sample 1:	492	/	4636		*9.8% – 11.5%*
Sample 2:	542	/	4534		*11% – 12.9%*

Verdict:

Sample 2 is more successful

(p = 0.0423)

Confidence level: 95%



Conclusion and recommendations

Next steps: As a result of testing, there was no positive dynamics in either the main or additional metrics. Adding information (`add_payment_info`) takes too much time to achieve a statistically significant result. We cannot recommend using these changes.